Materials and Methods Draft

Mouse DRG RNA-seq Follow-up Project

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Draft prepared for BIOL550

Materials and Methods

This project reanalyzed mouse DRG bulk RNA-seq after sciatic nerve injury using a four-stage workflow modeled after a CRISP-style progression: Data Collection, Data Cleaning, Data Preparation, and Data Analysis and Interpretation. That structure was not added only for presentation. It reflects how the mouse_new workflow was built and refined across the weekly reports, where each stage owned a specific handoff, a checkpoint artifact, and a reusable output for the next stage. The same staged structure also made it possible to reuse the pipeline after earlier dataset-selection problems, because accession tracking, QC review, alignment review, and downstream modeling were separated into explicit, repeatable checkpoints instead of being handled as one monolithic notebook run.

The working subset comprised 20 NovaSeq X paired-end libraries from SRP618841 (BioProject PRJNA1017789; GEO GSE243308), organized as a balanced side-by-genotype design before quality control, alignment, and differential expression modeling. In practice, the weekly reports served as the running record of how this dataset was narrowed, checked, and carried forward, while the final Methods draft translates that same path into manuscript prose.

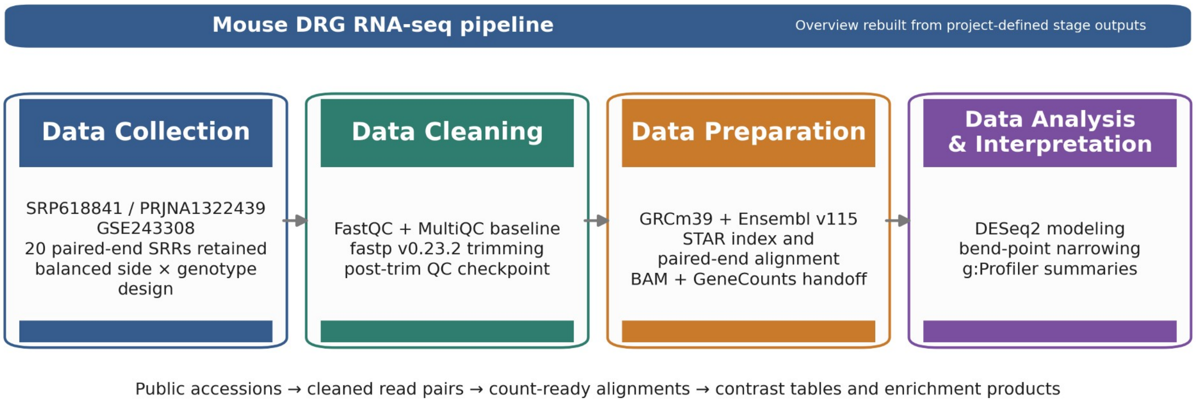

Figure 1. Overview ribbon summarizing the four computational stages and their stage-owned handoffs in the mouse_new

Table 1. Pipeline checkpoint table used to anchor the Methods section to stage-specific evidence.

Stage	Primary artifacts	Core tools	Checkpoint metric	Handoff
Collection	sample table; family sheet	prefetch + fasterq-dump	20 SRRs retained; 4 balanced subgroups (n=5 each)	FASTQ.gz pairs + manifest
Cleaning	QC heatmap; severity delta	FastQC + MultiQC + fastp	raw adapter-content failures resolved after trimming	trimmed read pairs plus per-sample QC reports
Preparation	unique-mapping plot; STAR median summary	STAR + GeneCounts	93.24% median unique mapping; 3.82% median noFeature	sorted BAMs, STAR logs, count tables, and family matrix
Analysis	filtering summary analysis summary g:Profiler source tables	DESeq2 + bend-point + g:Profiler	78,334 -> 21,481 genes after filtering; 5 modeled contrasts	DE tables, bend-point follow-up sets, and enrichment summaries

Representative command examples

The course draft requirements asked for sample commands for each major tool. the corresponding table therefore provides short representative examples showing how the main workflow stages were executed. These examples are included to document the computational path and major non-default parameters; they are illustrative rather than exhaustive wrapper scripts.

Table 2. Representative command examples for the tools used in the mouse_new

Stage	Tool	Representative		

		command / code example		
Collection	prefetch fasterq-dump	prefetch SRRXXXXXXX		fasterq-dump SRRXXXXXXX --split-files -O fastq/
Cleaning	FastQC	fastqc raw/*.fastq.gz trimmed/*.fastq.gz --outdir fastqc_reports/		
Cleaning	MultiQC	multiqc fastqc_reports/ fastp_reports_full/ -o multiqc/		
Cleaning	fastp	fastp -i sample_R1.fastq.gz -I sample_R2.fastq.gz -o clean_R1.fastq.gz -O clean_R2.fastq.gz --detect_adapter_for_pe --trim_poly_g --cut_mean_quality 20 --length_required 30 -h sample.html -j sample.json		
Preparation	STAR	STAR --runThreadN 8 --genomeDir star_index --readFilesIn clean_R1.fastq.gz clean_R2.fastq.gz --readFilesCommand zcat --sjdbGTFfile Mus_musculus.GRCm39.115.gtf --outSAMtype BAM SortedByCoordinate --quantMode GeneCounts		
Analysis	DESeq2	dds <- DESeqDataSetFromMatrix(countData, colData, design = side_class * geno_class); dds <- DESeq(dds)		
Analysis	bend-point scripts	python3 run_bendpoint_followup.py --contrast ipsi_vs_contra_in_ff --padj-column padj		
Analysis	g:Profiler	gp.profile(organism = "mmusculus", query = selected_genes, sources = c("GO:BP","KEGG","REAC"))		

Data Collection: SRA Acquisition and Sample Organization

The analysis began with the published DRG RNA-seq study SRP618841, linked to BioProject PRJNA1017789 and GEO series GSE243308. Public runs were retrieved using the project wrappers around prefetch and fasterq-dump, and the local workflow retained 20 paired-end NovaSeq X libraries from the 1-day-post-injury DRG subset. Recording this accession lineage was important because the pipeline had to remain reusable across candidate mouse datasets; explicit accession-to-run tracking prevented later stages from being tied to an ambiguous ``mouse'' source.

After download, metadata were consolidated into a design-aware sample table rather than being left as file-name strings. Each retained sample was annotated by side_class (ipsi or contra), geno_class (ff or cre), and the derived condition_family. This step established the balanced 22 design used in the DESeq2 model and in subsequent interpretation. It also created a reusable family manifest that could be re-read by downstream notebooks and figure builders without having to reconstruct group identity from filenames each time.

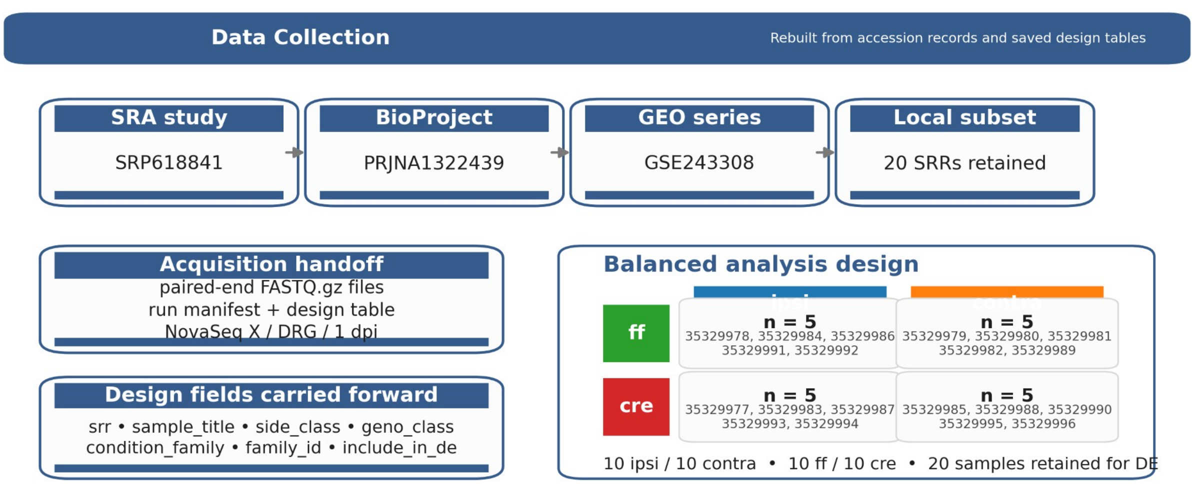

Figure 2. Rebuilt Data Collection stage figure derived from the saved project metadata and design tables.

Table 3. Data Collection stage table.

Substage	Input	Tool	Checkpoint artifact / evidence	Why it mattered
Study provenance	public accession records	SRA / GEO review	accession lineage shown in the corresponding figure: SRP618841 -> PRJNA1017789 -> GSE243308	anchors the Methods section to the exact source study
Run acquisition	public SRA runs	prefetch fasterq-dump	paired-end FASTQ.gz files plus the local run manifest; 20 retained SRRs from the DRG / NovaSeq X subset	creates the fixed read set used throughout the pipeline
Design mapping	run manifest + sample titles	local manifest assembly	design table and family sample sheet; 5 samples in each subgroup	makes the side/genotype structure explicit before PCA and DE modeling

Data Cleaning: Quality Control and Read Trimming

Raw read quality was evaluated with FastQC, and cohort-level summaries were consolidated with MultiQC. This stage was treated as a documented checkpoint rather than a minimal preprocessing step, so before-and-after QC artifacts were retained to show the effect of trimming explicitly. The local workflow used fastp v0.23.2 as the canonical cleanup tool because it provided a reproducible paired-end workflow, generated per-sample reports that could be reviewed later, and resolved adapter-related signal more effectively than the earlier baseline.

The retained fastp configuration used paired-end adapter detection, poly-G trimming, a mean-quality cutoff of 20, a minimum retained length of 30 bases, and per-sample HTML/JSON reporting. In the corresponding figure, the left panel shows the module-status heatmap and the right panel shows the SRR-level severity-change plot. Together, these artifacts document a consistent improvement in read quality across the cohort before alignment. Keeping both the raw and trimmed checkpoints made this stage reusable for later dataset screening, because poor candidate subsets could be rejected or repaired using the same evidence trail rather than by ad hoc trimming decisions.

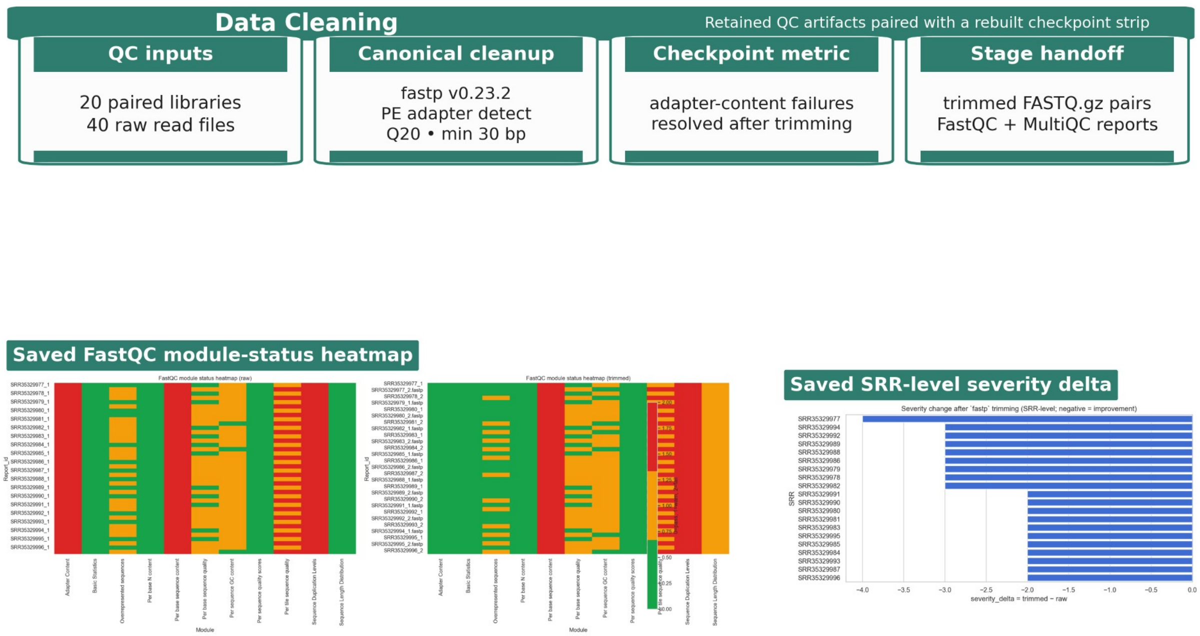

Figure 3. Data Cleaning stage figure using retained QC artifacts and a rebuilt checkpoint strip.

Table 4. Data Cleaning stage table.

Substage	Input	Tool	Checkpoint artifact / evidence	Why it mattered
Raw QC	raw paired-end reads	FastQC + MultiQC	raw module-status heatmap and cohort QC summary; adapter-content failures across all 40 raw read files	establishes the baseline quality state before trimming
Read trimming	raw FASTQ.gz pairs	fastp v0.23.2	trimmed read pairs per-sample HTML/JSON reports; PE adapter detect, Q20 cutoff, min length 30 bp, poly-G trimming	standardizes cleanup across all retained libraries
Post-trim checkpoint	trimmed read pairs	FastQC + MultiQC	saved SRR-level severity summary; all retained severity deltas moved in the negative direction after trimming	justifies using the cleaned reads, not the raw reads, for alignment

Data Preparation: Reference Selection and Alignment

Trimmed read pairs were aligned to the Mus musculus GRCm39 primary assembly with the matching Ensembl release 115 annotation. The reference pair used in the local workflow consisted of Mus_musculus.GRCm39.dna.primary_assembly.fa and Mus_musculus.GRCm39.115.gtf. A STAR genome index was generated with sjdbOverhang = 150 to match the 151-bp paired-end reads. Alignment was performed with STAR using gzip-aware input handling, coordinate-sorted BAM output, and GeneCounts so that the same run also produced per-sample count tables.

This stage linked cleaned reads to downstream modeling through a stable alignment handoff. For each sample, the workflow retained three key outputs: a coordinate-sorted BAM file, a STAR Log.final.out summary, and a ReadsPerGene.out.tab count table. These outputs provided both the alignment checkpoint and the count-level handoff used to assemble the final family matrix for the DRG subset. Treating the alignment review and the count-matrix assembly as explicit handoffs made it possible to reuse the same preparation stage across datasets without rewriting the downstream DE workflow.

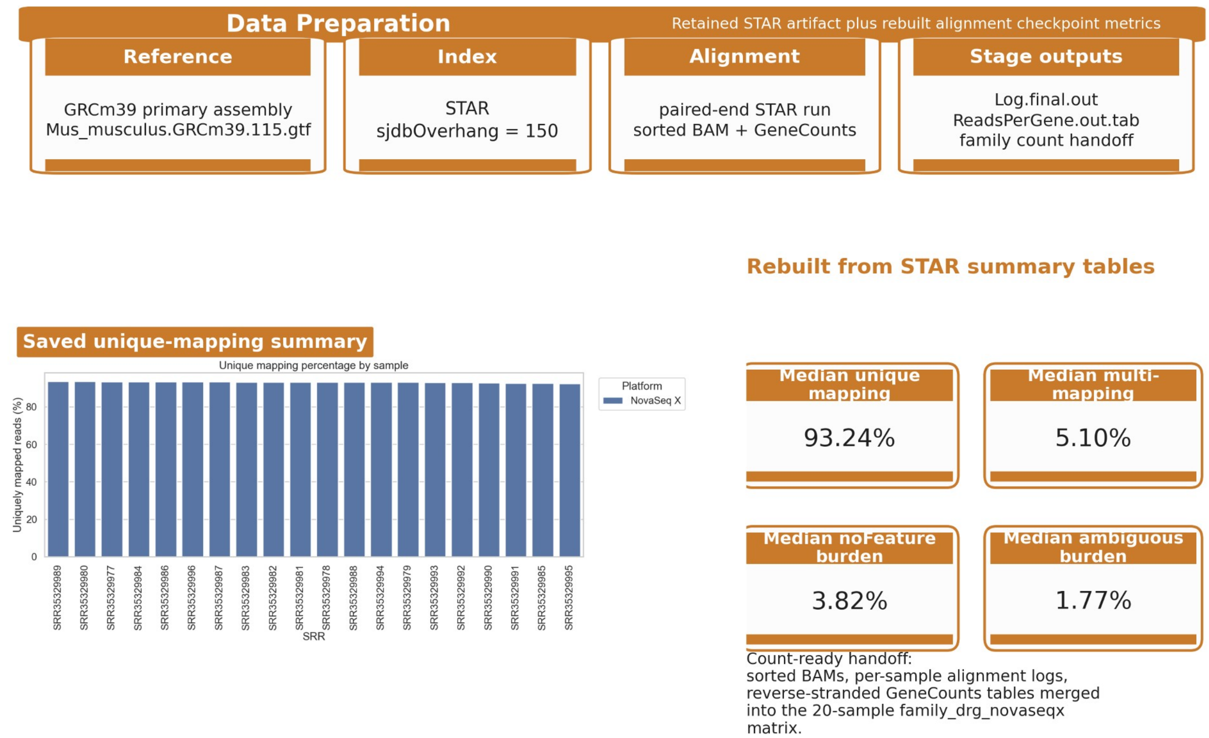

Figure 4. Data Preparation stage figure built from retained STAR outputs and rebuilt alignment checkpoint metrics.

Table 5. Data Preparation stage table.

Substage	Input	Tool	Checkpoint artifact / evidence	Why it mattered	
Reference choice	mouse genome + annotation	Ensembl reference selection	reference FASTA/GTF pair for GRCm39 primary assembly + Ensembl v115 annotation	locks the coordinate system used by STAR and downstream counts	
stageprepSTAR index \	run	trimmed read pairs + reference	STAR	sorted BAM, Log.final.out, and ReadsPerGene.out .tab; sjdbOverhang = 150 and paired-end alignment/counting in one run	produces the alignment checkpoint and the count-ready artifacts in one run
Alignment checkpoint	saved STAR summaries	local alignment review	retained unique-mapping plot plus the STAR median summary table; unique 93.24%, multi 5.10%, noFeature 3.82%, ambiguous 1.77%	confirms the cleaned reads mapped well enough to move into DESeq2	
Count handoff	per-sample count outputs	local merge scripts /	family-level count matrix for	converts per-sample STAR	

		notebooks	family_drg_novaseqx; 20 samples retained in the final family matrix	outputs into the DE-ready family matrix	

Data Analysis and Interpretation: Differential Expression and Functional Analysis

The merged family count matrix was analyzed with DESeq2 using the explicit side-by-genotype design carried forward from the sample manifest. Before modeling, low-support genes were filtered, reducing the tested set from 78,334 genes to 21,481 while retaining all 20 DRG samples. Five contrasts were then evaluated: ipsi_vs_contra_in_ff, ipsi_vs_contra_in_cre, geno_in_contra, geno_in_ipsi, and interaction. This contrast inventory was kept fixed so that later interpretation did not drift when one branch became more interpretable than another.

As documented in the weekly reports, downstream interpretation followed a PCA-first and branch-priority logic rather than a raw gene-count logic. PCA was used to read sample structure before long DE tables were emphasized, and the side-specific branches were then carried forward as the primary follow-up path. To avoid reducing very large DE outputs to unstructured gene lists, the workflow applied a bend-point rule to the ordered adjusted-p-value curve for each contrast. These bend-point-derived follow-up sets were therefore reported alongside the padj < 0.05 results rather than treated as interchangeable with them. For the two primary side-specific contrasts, this step retained 709 and 870 genes, respectively.

The resulting follow-up sets were then submitted to g:Profiler using GO Biological Process, KEGG, and Reactome layers. In the ff branch, which remained the clearest pathway-level follow-up in the weekly reports, enrichment outputs were later re-read through shared-gene chart, tree, and network views to reduce redundancy among overlapping GO labels. In this

Methods framing, those enrichment products are described as reusable downstream summaries of the DE workflow rather than as final biological claims.

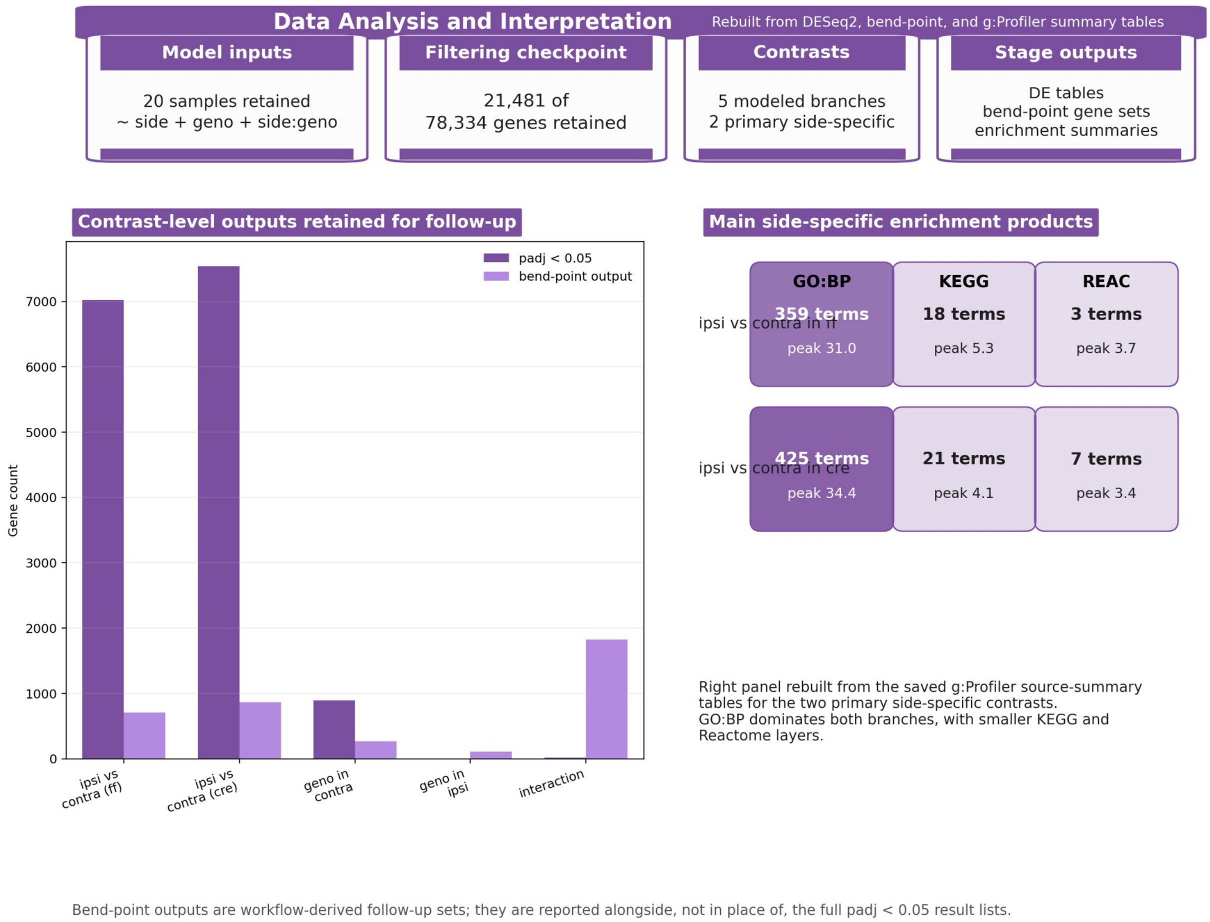

Figure 5. Rebuilt Data Analysis and Interpretation stage figure derived from saved DESeq2

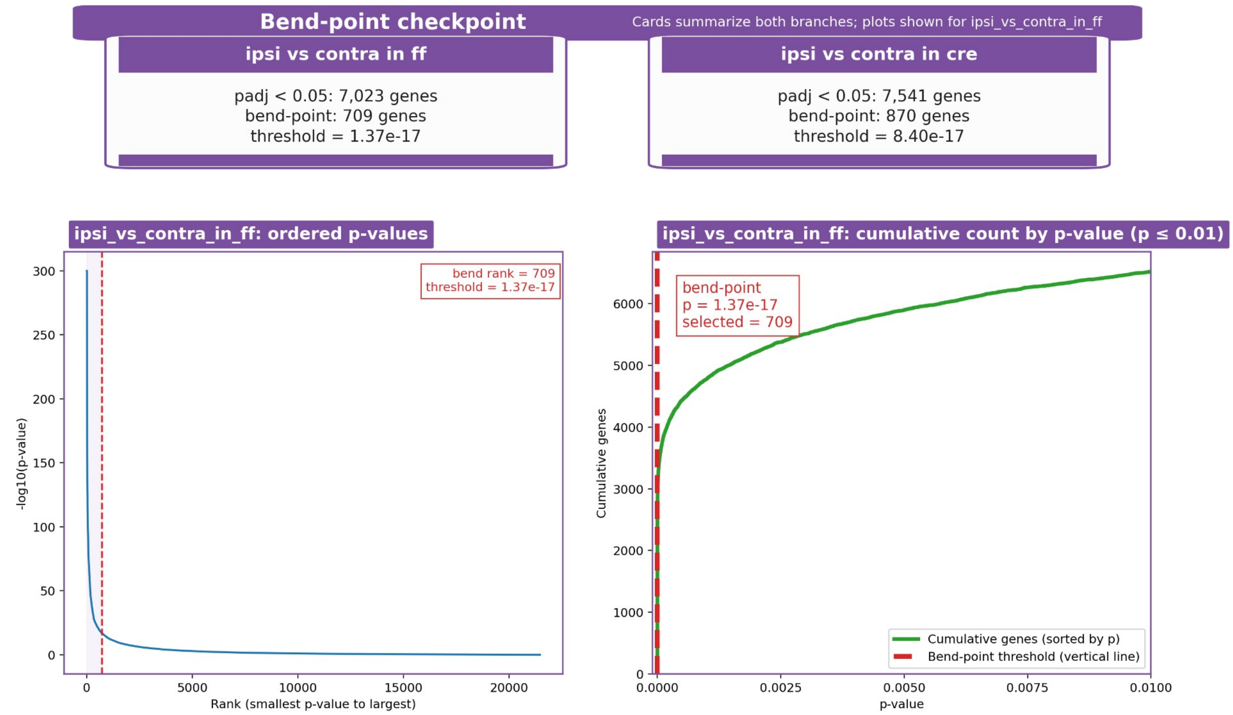

Figure 6. Retained bend-point checkpoint summary for the two primary side-specific branches (cards). The ordered p-value and cumulative-count plots are shown for the ipsi_vs_contra_in_ff

Table 6. Data Analysis and Interpretation workflow table.

Substage	Input	Tool	Checkpoint artifact / evidence	Why it mattered
Model setup	family count matrix + design table	DESeq2	normalized counts, VST matrix, and contrast-level result tables; 20 samples retained and 78,334 -> 21,481 genes after filtering	defines the inferential base for every contrast and downstream plot
Contrast production	filtered model outputs	DESeq2 contrast manifest	five saved contrast branches; two primary side-specific and three supporting genotype / interaction branches	keeps the main and supporting stories explicit before list narrowing
Bend-point selection	ordered adjusted-p tables	local bend-point scripts	analysis-summary and per-contrast bend-point summary tables; primary side-specific follow-up sets of 709 genes in ff and 870 genes in cre	turns very large DE lists into smaller reproducible follow-up sets
Enrichment layer	bend-point-selected gene sets	g:Profiler	g:Profiler source-summary and enrichment tables; the main ff branch retained 359 GO:BP, 18 KEGG, and 3 Reactome terms	captures the terminal functional-summary products without replacing the gene-level story

Table 7. Primary side-specific follow-up summary carried out during the Data Analysis stage.

Branch	padj < 0.05 genes	Bend-point genes	Threshold	GO:BP	KEGG	REAC
ff side branch	7,023	709	1.37e-17	359	18	3
cre side branch	7,541	870	8.40e-17	425	21	7